

# PQC Cryptographic Inventory

2026, 2027 and beyond

Version 2.0 | Reviewed 29 September 2026

Prepared by Ankit Gupta

## Static PDF companion

Use the matching XLSX for editable inventory records, formulas, drop-downs, evidence, owners and status. This PDF is a readable reference and is not an interactive workbook.

## How to use this companion

Start with scope and accountable owners. Record the cryptographic dependency, confidentiality or authenticity lifetime, algorithm and implementation evidence. Prioritize exposed, long-lived, high-impact dependencies; confirm supported protocol and product paths; test interoperability, failure modes, rollback and operational impact before deployment.

Final standards, drafts, research selections and local planning targets are different statuses. NIST alignment means the material draws from the listed sources; it does not claim NIST endorsement or completeness of every future recommendation.

Workbook: [secureazcloud-pqc-crypto-inventory-template-v2.0.xlsx](#)

## Included worksheets

Start Here | Crypto Inventory | PQC Readiness Checklist | Cloud IAM Playbook | SCADA ICS Checklist | References | Roadmap | Threat Register

# Start Here

Read this section alongside the corresponding worksheet in the editable workbook. Record decisions and evidence there.

NIST publication status checked 2026-09-29. Forward planning covers 2026, 2027 and later review cycles. Examples only; replace with evidence from your environment.

## Use this workbook

- 1: Inventory** | Create one record per cryptographic component/function; link parent and peer IDs. Include transport, stored data, signatures, firmware, backups and plaintext boundaries.
- 2: Classify risk** | Record evidence, confidentiality and signature trust lifetimes, criticality and agility. Unknown inputs remain unscored. This is a local screening heuristic.
- 3: Plan migration** | Assign migration owners, supported versions, dependencies, product EOL, milestones and expiring exceptions. Use Roadmap and Threat Register.
- 4: Pilot and validate** | Verify exact parameter/profile, end-to-end compatibility, implementation security, performance and recovery before production approval.
- 5: Operationalize** | Review sources and suppliers quarterly and on material change. Monitor draft-to-final changes, errata and cryptanalysis. The cadence is local guidance.

## Navigation and inventory summary

**Crypto Inventory:** Master asset-level crypto inventory with risk scoring and migration priority.

**PQC Readiness Checklist:** Program readiness checklist mapped to NIST CSF-style outcomes and NIST PQC migration concepts.

**Cloud IAM Playbook:** Phase-based cloud/IAM migration planning checklist for identity, PKI, TLS, federation, workload identities, and cryptographic dependencies.

**SCADA ICS Checklist:** Safety-first OT/ICS/SCADA readiness checklist for cryptographic discovery, remote access, vendor roadmaps, and change control.

**References:** Public source links and alignment notes.

**Lists:** Dropdown values used by the templates.

**Roadmap:** Local 2026, 2027, 2028-2030, 2031-2035 and ongoing actions.

**Threat Register:** Assess threats by asset with owner, due date and evidence.

**Method / limits:** The 0-100 score and P0-P4 bands are SecureAzCloud screening choices, not a NIST score, compliance result or prediction of Q-day.

**Score weights:** Vulnerable 20; criticality  $1-5 \times 5$ ; HNDL 15; exposure public/partner/internal 10/7/3; hard-coded/configurable 10/3; max(confidentiality, signing life)  $\geq 10/5/2$  years 10/7/3; safety high/medium 10/5.

**Score bands:**  $P0 \geq 70$ ;  $P1 \geq 50$ ;  $P2 \geq 30$ ;  $P3 \geq 15$ ;  $P4 < 15$ . Blank, unknown or invalid scoring inputs show Incomplete / Review inputs. Zero lifetime is a valid input.

**Required evidence:** Actual rows require owners, source/date, provenance, confidence, function, dependency IDs and review date. Evidence Completeness is a data check, not security approval.

**Risk override:** Use Priority Override / Rationale for justified owner decisions. High consequence or long lead time can justify earlier action regardless of score. Never store raw keys or secrets.

**Final standards:** FIPS 203/204/205, SP 800-227 and CSWP 39-upd1 are final. KEM and signatures serve different functions; a PQC transport does not migrate token or certificate signatures.

**Draft timing:** IR 8547 is draft. It proposes deprecation after 2030 for 112-bit uses, and disallowance after 2035 for listed quantum-vulnerable uses. It is not a blanket 2030 mandate or universal legal deadline.

**2027 and later:** HQC/FN-DSA remain under standardization; new signatures are candidates. Future unpublished guidance cannot be pre-certified. Recheck final/draft status and errata before deployment.

**Completion gate:** Mark Complete only after approved profile, vendor support, end-to-end tests, key recovery, observed deployment, owner acceptance and documented residual risk.

**Field use:** A:AH retain the original inventory. AI:AN establish provenance and function. AO:AT capture strength, lifetimes and dependencies. AU:BN capture target profile, validation, lifecycle, exceptions and evidence.

**Version / date:** v2.0 / 2026-09-29. Supersedes the unversioned v1.0 inventory pack. Keep original examples flagged Example and add Actual records for your environment.

**Author / Maintainer:** Ankit Gupta

# Crypto Inventory

The editable workbook contains the full data-entry template. The field structure below identifies information to collect. Example records are illustrative and are not a validated production inventory.

## Fields in the editable worksheet

01. Asset ID
02. Asset/System Name
03. Business Process
04. Business Owner
05. Technical Owner
06. Environment
07. Location/Zone
08. Data Classification
09. Confidentiality Life (years)
10. External Exposure
11. OT/ICS Safety Impact
12. Criticality (1-5)
13. Crypto Use Case
14. Protocol/Standard
15. Algorithm Family
16. Algorithm/Curve/Group
17. Key Size (bits) / Cert Bytes
18. Crypto Library / HSM / CA
19. Certificate/Key Location
20. Identity Type
21. Vendor/Supplier
22. Vendor PQC Roadmap
23. Crypto Agility
24. PQC Target Standard
25. Migration Pattern
26. Quantum Vulnerable
27. Harvest-Now Concern
28. Local Risk Score (0-100)
29. Local Migration Priority
30. Status
31. Target Quarter
32. Evidence URL
33. Last Verified Date
34. Notes
35. Record Type
36. Discovery Method
37. Discovery Evidence / Scope
38. Discovery Date
39. Evidence Confidence
40. Crypto Function
41. Classical Strength (bits)
42. Signature Trust Life (years)
43. Component / Provider Version

- 44. CBOM / Dependency IDs
- 45. Parent / Peer Record IDs
- 46. Termination / Plaintext Legs
- 47. Target Parameter Set
- 48. Protocol Profile / Readiness
- 49. Interop / Performance Evidence
- 50. Algorithm CAVP Evidence
- 51. CMVP Module / Version / Scope
- 52. Product End of Support
- 53. Migration Lead Time (months)
- 54. Migration Owner
- 55. Next Review Date
- 56. Target Completion Date
- 57. Exception ID
- 58. Exception Approver / Expiry
- 59. Residual Risk / Treatment
- 60. Key Lifecycle / Backup Recovery
- 61. RNG / Key Generation Evidence
- 62. Implementation Security Evidence
- 63. Signing / Archive Transition
- 64. Downgrade / Fallback Policy
- 65. Evidence Completeness
- 66. Priority Override / Rationale

## Illustrative records - selected fields

### 01 Customer-facing web portal

**Asset ID:** APP-001

**Business Process:** Customer access

**Crypto Use Case:** PKI/Certificate

**Protocol/Standard:** TLS certificate authentication

**Algorithm Family:** RSA

**Algorithm/Curve/Group:** RSA-2048 signing

**PQC Target Standard:** ML-DSA

**Target Quarter:** 2026-Q4

**Notes:** EXAMPLE ONLY. Signing component. Transport key establishment is APP-001-KEX. Vendor/protocol support must be verified.

**Crypto Function:** Digital signature

**Target Parameter Set:** TBD after profile selection

## 02 Enterprise SSO signing certificate

**Asset ID:** IAM-001

**Business Process:** Workforce authentication

**Crypto Use Case:** SAML

**Protocol/Standard:** SAML 2.0 signing

**Algorithm Family:** RSA

**Algorithm/Curve/Group:** RSA-3072 signing cert

**PQC Target Standard:** ML-DSA

**Target Quarter:** 2026-Q3

**Notes:** EXAMPLE ONLY. Partner support and validator profile govern any signing change. No actual product support claimed.

**Crypto Function:** Digital signature

**Target Parameter Set:** TBD after profile selection

# PQC Readiness Checklist

Use each item with evidence, an accountable owner, a target date, and the editable workbook status fields. An unchecked control is not satisfied by this reference document.

## 01 Assign an accountable PQC migration owner and cross-functional working group.

**Domain:** Governance

**Acceptance Criteria:** Named owner, RACI, recurring cadence, decision log.

**Evidence / Artifact:** Program charter, RACI, meeting notes.

**Suggested Owner:** Security leadership / architecture

**NIST Alignment:** NIST CSF 2.0 Govern; NCCoE Migration to PQC

## 02 Define enterprise crypto-agility policy.

**Domain:** Governance

**Acceptance Criteria:** Policy requires configurable algorithms, approved cryptographic libraries, and migration-friendly architectures.

**Evidence / Artifact:** Policy document, architecture standard.

**Suggested Owner:** Security architecture

**NIST Alignment:** N05 CSWP 39-upd1 final

## 03 Build a cryptographic inventory across applications, services, devices, certificates, libraries, protocols, and identities.

**Domain:** Inventory

**Acceptance Criteria:** Inventory covers at least externally exposed, high-value, long-lived, regulated, and OT/ICS systems.

**Evidence / Artifact:** Completed Crypto Inventory sheet.

**Suggested Owner:** Security engineering / asset owners

**NIST Alignment:** NCCoE Migration to PQC; CISA quantum-readiness roadmap

## 04 Identify store-now/decrypt-later exposure.

**Domain:** Data risk

**Acceptance Criteria:** Data shelf life and sensitivity are recorded; long-life confidential data is prioritized.

**Evidence / Artifact:** Data classification and retention mapping.

**Suggested Owner:** Data governance / privacy / security

**NIST Alignment:** NIST PQC migration planning

## 05 Classify crypto agility for each asset.

**Domain:** Architecture

**Acceptance Criteria:** Hard-coded algorithms and non-upgradeable libraries are flagged as migration blockers.

**Evidence / Artifact:** Crypto agility field in inventory.

**Suggested Owner:** Application owners / platform teams

**NIST Alignment:** NIST CSWP 39

## 06 Assess certificate authorities, certificate profiles, issuing workflows, trust anchors, and renewal automation.

**Domain:** PKI

**Acceptance Criteria:** PKI roadmap includes algorithm agility, chain size impact, relying-party testing, rollback, and monitoring.

**Evidence / Artifact:** PKI assessment, CA roadmap, certificate profile matrix.

**Suggested Owner:** PKI/IAM team

**NIST Alignment:** FIPS 204; FIPS 205; SP 800-53 IA/SC

## 07 Map authentication and federation dependencies using public-key signatures.

**Domain:** Cloud/IAM

**Acceptance Criteria:** SAML/OIDC/OAuth, service identities, workload identities, SSH, device identities, and signing flows are inventoried.

**Evidence / Artifact:** Cloud/IAM playbook output.

**Suggested Owner:** IAM / cloud platform team

**NIST Alignment:** CSF 2.0 Govern/Identify/Protect; FIPS 204/205

## 08 Identify TLS, VPN, SSH, API gateway, service mesh, and remote access dependencies.

**Domain:** Network

**Acceptance Criteria:** Externally exposed and partner-facing endpoints have migration readiness owners and test environments.

**Evidence / Artifact:** Endpoint and protocol inventory.

**Suggested Owner:** Network / platform security

**NIST Alignment:** FIPS 203; NCCoE Migration to PQC

## 09 Request vendor PQC roadmaps and product support statements.

**Domain:** Suppliers

**Acceptance Criteria:** Critical vendors provide PQC roadmap, dependency disclosures, update mechanisms, and support windows.

**Evidence / Artifact:** Vendor questionnaire responses, contracts, SLAs.

**Suggested Owner:** Procurement / vendor risk

**NIST Alignment:** SP 800-161r1; CISA product categories

## 10 Create PQC interoperability and performance test plan.

**Domain:** Testing

**Acceptance Criteria:** Test plan includes certificate size, handshake behavior, latency, logging, monitoring, failure modes, and rollback.

**Evidence / Artifact:** Test plan, lab results.

**Suggested Owner:** Engineering / QA / security

**NIST Alignment:** NCCoE interoperability workstream; NIST CSWP 39

## 11 Use safety-first OT/ICS change governance.

**Domain:** OT/ICS

**Acceptance Criteria:** No production OT crypto changes occur without safety impact assessment, vendor validation, lab testing, MOC approval, and maintenance window.

**Evidence / Artifact:** MOC records, lab evidence, vendor approval.

**Suggested Owner:** OT security / operations

**NIST Alignment:** NIST SP 800-82r3; CISA OT PQC considerations

## 12 Update monitoring, incident response, and audit evidence for changed cryptographic protocols and certificates.

**Domain:** Operations

**Acceptance Criteria:** Logs and alerts validate new certificates/protocols; runbooks reflect rollback and exception handling.

**Evidence / Artifact:** Monitoring rules, runbooks, change records.

**Suggested Owner:** SOC / operations

**NIST Alignment:** SP 800-53 AU/IR/SC; CSF Detect/Respond/Recover

## 13 Publish role-based awareness for engineers, asset owners, and procurement.

**Domain:** Training

**Acceptance Criteria:** Audience-specific guidance explains inventory fields, vendor questions, testing expectations, and exception process.

**Evidence / Artifact:** Training materials, communications.

**Suggested Owner:** Security enablement

**NIST Alignment:** CSF Govern; SP 800-53 AT

## 14 Separate cryptographic functions and relationship records.

**Domain:** Inventory

**Acceptance Criteria:** Record key establishment, signatures, key wrapping, symmetric protection and hashes separately. Reconcile CBOM, code/configuration and observed network evidence.

**Evidence / Artifact:** Component IDs, dependency/peer mapping and discovery coverage.

**Suggested Owner:** Security engineering

**NIST Alignment:** N05 final; N07 project; N08 preliminary draft

**Status:** Not assessed

**Notes:** Algorithm discovery is not proof of whole-system security.

## 15 Separate confidentiality and signature risks.

**Domain:** Threats

**Acceptance Criteria:** Record confidentiality lifetime, long-lived verification obligations and migration lead time. Assess HNDL for confidentiality paths and forgery/impersonation for signing paths.

**Evidence / Artifact:** Data retention and signature trust horizon.

**Suggested Owner:** Data owners / PKI

**NIST Alignment:** N06 draft; N09 final

**Status:** Not assessed

**Notes:** No assumed Q-day. Assess signing keys, firmware and immutable trust anchors.

## 16 Choose standards and exact parameter/profile.

**Domain:** Cryptography

**Acceptance Criteria:** Use FIPS 203 for KEM, FIPS 204/205 for signatures. Record exact parameters and supported integration. ML-KEM does not itself encrypt application data or authenticate peers.

**Evidence / Artifact:** Approved profile, protocol, parameter set and vendor version.

**Suggested Owner:** Crypto architecture

**NIST Alignment:** N01-N04 final

**Status:** Not assessed

**Notes:** HQC/FN-DSA and additional signatures are watch items until standardized.

## 17 Verify KEM and implementation security.

**Domain:** Implementation

**Acceptance Criteria:** Test input validation, randomness, key derivation, authentication, key confirmation when required, secret handling/destruction and side-channel resistance. Use supported constructions.

**Evidence / Artifact:** Implementation review and negative-test evidence.

**Suggested Owner:** Security engineering

**NIST Alignment:** N04 final

**Status:** Not assessed

**Notes:** No home-built hybrid combiners.

## 18 Separate algorithm testing from module validation.

**Domain:** Validation

**Acceptance Criteria:** Record algorithm test evidence separately from module certificate, exact software/firmware version, operational environment, approved mode, security policy, PQC service scope and current status/date where validation is required.

**Evidence / Artifact:** CAVP evidence and CMVP module validation record.

**Suggested Owner:** GRC / platform owner

**NIST Alignment:** N24-N26 official programs

**Status:** Not assessed

**Notes:** A FIPS algorithm name or CAVP certificate is not product-wide FIPS 140 validation. Historical module status is not revoked status.

## 19 Track key and verification lifecycle.

**Domain:** Lifecycle

**Acceptance Criteria:** Include issuance, protection, rotation, revocation, recovery, backup, destruction, trust-store distribution and archived signature verification obligations.

**Evidence / Artifact:** Key lifecycle policy and restore/rollover tests.

**Suggested Owner:** PKI / key management

**NIST Alignment:** N09 final; N10 draft watch

**Status:** Not assessed

**Notes:** Store key references only, never private keys, tokens or secrets in this workbook.

## 20 Tie migration to supported product releases and retirement.

**Domain:** Supplier

**Acceptance Criteria:** Obtain exact release/version commitments, end-of-support, HSM/KMS/CA dependencies, support constraints and lead time. Budget replacement for immutable verifiers.

**Evidence / Artifact:** Supplier evidence, roadmap owner, funded milestones.

**Suggested Owner:** Procurement / asset owner

**NIST Alignment:** N05 final

**Status:** Not assessed

**Notes:** Vendor marketing alone is insufficient evidence.

## 21 Require end-to-end acceptance before Complete.

**Domain:** Assurance

**Acceptance Criteria:** All path components and peers meet approved security profile, interoperability, performance, rollback and evidence requirements. Close residual risks or approve time-bound exceptions.

**Evidence / Artifact:** Signed acceptance, observed configuration, test report.

**Suggested Owner:** Service / risk owners

**NIST Alignment:** N04-N05 final; N07 project

**Status:** Not assessed

**Notes:** Local score and checklist status are not NIST certification.

## 22 Operate standards and errata review.

**Domain:** Standards

**Acceptance Criteria:** Review quarterly and upon errata, cryptanalysis, final standards, module-status change, releases and incidents. Reassess targets against current authoritative publications.

**Evidence / Artifact:** Dated review log and changed decisions.

**Suggested Owner:** Crypto governance

**NIST Alignment:** N01-N06; N10/N12/N14 draft watch; N19-N21

**Status:** Not assessed

**Notes:** Quarterly cadence is a local recommendation. Future unpublished 2027+ guidance is not pre-certified.

## 23 Track risk-based phases beyond 2027.

**Domain:** Roadmap

**Acceptance Criteria:** Use the Roadmap sheet for local 2026, 2027, 2028-2030, 2031-2035 and ongoing milestones; apply earlier organizational or sector requirements where applicable.

**Evidence / Artifact:** Owners, due dates, dependencies, exceptions and closure evidence.

**Suggested Owner:** Program owner

**NIST Alignment:** N05 final; N06 draft

**Status:** Not assessed

**Notes:** IR 8547 proposes after-2030 deprecation for 112-bit uses and after-2035 disallowance for listed vulnerable uses. It remains draft.

## 24 Recheck the 2026 FIPS module transition.

**Domain:** Validation

**Acceptance Criteria:** Review actual module certificate and applicability. FIPS 140-2 modules moved from the active list after 2026-09-21. Confirm status, approved services and operational environment before deployment.

**Evidence / Artifact:** Current certificate/status/date and deployment acceptance.

**Suggested Owner:** GRC / platform owner

**NIST Alignment:** N25-N26 official CMVP guidance

**Status:** Not assessed

**Notes:** Historical does not mean revoked. An embedded validated module does not validate an entire product or protocol.

## 25 Track 2026 key generation and limited-use signature drafts.

**Domain:** Standards watch

**Acceptance Criteria:** Monitor SP 800-133r3 PQC key generation/seed/HSM proposals and SP 800-230 limited-signature SLH-DSA proposals before adopting new options.

**Evidence / Artifact:** Dated standards watch and approved implementation changes.

**Suggested Owner:** Crypto governance

**NIST Alignment:** N30-N31 initial public drafts 2026

**Status:** Not assessed

**Notes:** Drafts do not create final approved parameter sets or requirements.

# Cloud IAM Playbook

Read this section alongside the corresponding worksheet in the editable workbook. Record decisions and evidence there.

## 01 Scope cloud and identity systems

**Phase:** 0. Mobilize

**Detailed Action:** Define scope for public, private, hybrid cloud, identity providers, directories, PKI, KMS/HSM, certificate automation, workload identities, federation, service mesh, API gateways, VPN, SSH, CI/CD, artifact signing, and secrets management.

**Output / Decision:** Approved scope and architecture map.

**Owner:** Cloud security lead

**Dependencies:** Executive sponsor, asset inventory

## 02 Define policy guardrails

**Phase:** 0. Mobilize

**Detailed Action:** Require approved cryptographic libraries, centralized certificate lifecycle, algorithm agility, supplier disclosure, and documented exceptions.

**Output / Decision:** Updated cloud/IAM cryptography standard.

**Owner:** Security architecture

**Dependencies:** Policy team, legal/procurement

## 03 Inventory machine identities and certificates

**Phase:** 1. Discover

**Detailed Action:** Export certificate inventory from CAs, endpoints, ingress gateways, API gateways, service mesh, VPN, SSH, repositories, CI/CD, and device trust stores.

**Output / Decision:** Machine identity and certificate inventory.

**Owner:** PKI/IAM team

**Dependencies:** Access to CA and platform APIs

## 04 Map federation and token-signing cryptography

**Phase:** 1. Discover

**Detailed Action:** Record SAML signing/encryption certs, OIDC/OAuth token signing keys, JWT algorithms, federation metadata, partner trust anchors, and rotation procedures.

**Output / Decision:** Federation crypto dependency map.

**Owner:** IAM team

**Dependencies:** Partner metadata, IdP configuration

## 05 Map transport and key-establishment paths

**Phase:** 1. Discover

**Detailed Action:** Document TLS, mTLS, SSH, VPN, service-to-service encryption, database drivers, and API gateway policies.

**Output / Decision:** Transport crypto inventory.

**Owner:** Network/platform teams

**Dependencies:** Traffic flow maps, discovery tooling

## 06 Score migration priority

**Phase:** 2. Prioritize

**Detailed Action:** Use quantum vulnerability, data shelf life, public exposure, criticality, crypto agility, and supplier readiness to set P0/P1/P2 priorities.

**Output / Decision:** Risk-scored backlog.

**Owner:** Security architecture

**Dependencies:** Crypto Inventory sheet

## 07 Identify blockers

**Phase:** 2. Prioritize

**Detailed Action:** Flag hard-coded algorithms, legacy dependencies, non-upgradeable appliances, unmanaged certificates, and supplier roadmap gaps.

**Output / Decision:** Blocker register and mitigation owners.

**Owner:** Architecture review board

**Dependencies:** Supplier responses, engineering input

## 08 Set target architecture

**Phase:** 3. Design

**Detailed Action:** Choose migration patterns: library abstraction, certificate lifecycle automation, protocol upgrade, hybrid capable key establishment where supported, hardware refresh, or compensating control.

**Output / Decision:** Target-state design and standards.

**Owner:** Cloud/IAM architects

**Dependencies:** NIST FIPS 203/204/205; protocol maturity

## 09 Design certificate and key lifecycle

**Phase:** 3. Design

**Detailed Action:** Plan certificate issuance, renewal, revocation, telemetry, staged rollout, and emergency rollback. Include chain-size and relying-party compatibility testing.

**Output / Decision:** PKI transition plan.

**Owner:** PKI team

**Dependencies:** CA, platform, partner dependencies

## 10 Define non-production pilot

**Phase:** 3. Design

**Detailed Action:** Select low-blast-radius services with representative TLS, federation, workload identity, signing, and monitoring dependencies.

**Output / Decision:** Pilot plan and success criteria.

**Owner:** Engineering lead

**Dependencies:** Test environment, observability

## 11 Run interoperability and performance tests

**Phase:** 4. Pilot

**Detailed Action:** Measure handshake behavior, certificate size, token validation, latency, logging, monitoring, error handling, and rollback.

**Output / Decision:** Pilot test report.

**Owner:** Engineering / QA / SRE

**Dependencies:** Load testing, client diversity

## 12 Validate security controls

**Phase:** 4. Pilot

**Detailed Action:** Confirm policy enforcement, key protection, audit logging, access control, certificate transparency/telemetry, and incident response runbooks.

**Output / Decision:** Control validation evidence.

**Owner:** Security operations

**Dependencies:** SOC, GRC, platform owners

## 13 Implement phased migration

**Phase:** 5. Rollout

**Detailed Action:** Roll out by priority cohort with maintenance windows, canary deployments, automated checks, and rollback triggers.

**Output / Decision:** Change records and migration status.

**Owner:** Platform/service owners

**Dependencies:** CAB/MOC approval

## 14 Coordinate partners

**Phase:** 5. Rollout

**Detailed Action:** Notify relying parties and partners of certificate/signing/protocol changes; obtain compatibility confirmation.

**Output / Decision:** Partner compatibility approvals.

**Owner:** IAM/vendor/partner management

**Dependencies:** Partner contacts, contracts

## 15 Update continuous discovery

**Phase:** 6. Operate

**Detailed Action:** Schedule recurring crypto inventory refresh from endpoints, certificates, code repos, containers, libraries, and network telemetry.

**Output / Decision:** Continuous inventory feed.

**Owner:** Security engineering

**Dependencies:** CMDB, scanners, CA logs, SBOMs

## 16 Track exceptions and residual risk

**Phase:** 6. Operate

**Detailed Action:** Document non-migrated assets, compensating controls, expiry dates, and accountable owners.

**Output / Decision:** Exception register and review cadence.

**Owner:** Risk owner

**Dependencies:** GRC workflow

## 17 Inventory complete issuer-to-validator graph

**Phase:** 1. Discover

**Detailed Action:** Map OIDC/OAuth/SAML, JWT/JOSE/COSE, JWKS, metadata, sidecars, gateways, SDKs and downstream validators. Include automated workload identities where present.

**Output / Decision:** Key use, issuer/validator ownership and trust distribution map.

**Owner:** IAM / platform teams

**Dependencies:** N16 IR 8587 final; N17 final

**Status:** Not assessed

**Notes:** Hybrid TLS does not migrate token/certificate signatures.

## 18 Gate identity signing changes by profile support

**Phase:** 3. Design

**Detailed Action:** Verify approved algorithm allowlists, issuer/audience binding, key-use restrictions, signed metadata, trust-anchor distribution and protocol/vendor maturity.

**Output / Decision:** Approved signing profile and every-validator compatibility.

**Owner:** IAM architecture

**Dependencies:** N16 final §7.7; N02-N03 final

**Status:** Not assessed

**Notes:** Do not insert unsupported PQC algorithms into existing tokens.

## 19 Test token and key rollover boundaries

**Phase:** 4. Pilot

**Detailed Action:** Test token lifetime/refresh, key overlap, JWKS cache expiry, replay protection, clock tolerance, revocation and compromise recovery.

**Output / Decision:** Positive/negative validation and recovery tests.

**Owner:** IAM / QA / SOC

**Dependencies:** N16 final

**Status:** Not assessed

**Notes:** Record measured product limits and failure behavior.

## 20 Measure larger object and traffic impact

**Phase:** 4. Pilot

**Detailed Action:** Test headers, cookies, proxies, brokers, parser limits, certificate chains, MTU/fragmentation, retries, CPU, throughput and p95/p99 latency.

**Output / Decision:** End-to-end load/interop evidence and approved thresholds.

**Owner:** SRE / QA

**Dependencies:** N04-N05 final; N07 project

**Status:** Not assessed

**Notes:** No assumed universal performance threshold.

## 21 Map transport termination and envelope encryption

**Phase:** 3. Design

**Detailed Action:** Record external/internal TLS legs, mesh hops, plaintext boundaries, database clients, storage/backup key wrapping and KMS/HSM roles.

**Output / Decision:** Distinct key establishment, signature and data-encryption components.

**Owner:** Platform / data security

**Dependencies:** N04,N09,N17,N22,N23 final

**Status:** Not assessed

**Notes:** AES data encryption does not remove vulnerable public-key key-wrapping dependencies.

## 22 Control fallback, negotiation and emergency substitution

**Phase:** 6. Operate

**Detailed Action:** Test downgrade resistance and configuration enforcement; document time-bound rollback authorization, monitoring and re-migration triggers.

**Output / Decision:** Policy evidence, rollback rehearsal and dated risk acceptance.

**Owner:** Security / service owner

**Dependencies:** N04-N05 final

**Status:** Not assessed

**Notes:** No silent fallback to an unapproved profile.

# SCADA ICS Checklist

Use each item with evidence, an accountable owner, a target date, and the editable workbook status fields. An unchecked control is not satisfied by this reference document.

## 01 Establish OT PQC change-control policy.

**Area:** Governance

**Safety / Operational Consideration:** No production change without safety impact assessment, management of change approval, vendor support, rollback plan, and approved maintenance window.

**Evidence / Artifact:** MOC procedure, safety approval template.

**Owner:** OT governance

**NIST Alignment:** NIST SP 800-82r3; CSF Govern

## 02 Inventory OT crypto use without disrupting operations.

**Area:** Inventory

**Safety / Operational Consideration:** Use passive discovery, configuration review, vendor docs, certificate exports, and engineering workstation review before active scanning.

**Evidence / Artifact:** OT crypto inventory, discovery method log.

**Owner:** OT security / engineering

**NIST Alignment:** NIST SP 800-82r3; NCCoE Migration to PQC

## 03 Classify OT assets by process criticality and safety impact.

**Area:** Inventory

**Safety / Operational Consideration:** Prioritize assets by process function, safety consequence, downtime tolerance, vendor support, and data sensitivity.

**Evidence / Artifact:** Criticality and safety mapping.

**Owner:** Operations / engineering

**NIST Alignment:** NIST SP 800-82r3 risk management

## 04 Document encrypted remote access paths.

**Area:** Network

**Safety / Operational Consideration:** Include VPN, jump hosts, vendor support links, modems, cloud connectors, remote engineering, bastions, and MFA dependencies.

**Evidence / Artifact:** Remote access architecture map.

**Owner:** OT network team

**NIST Alignment:** SP 800-82r3; SP 800-53 AC/IA/SC

## 05 Review industrial protocol security settings.

**Area:** Network

**Safety / Operational Consideration:** Identify where TLS, certificates, signed firmware, secure boot, or application-layer security are enabled, unsupported, or vendor-dependent.

**Evidence / Artifact:** Protocol security matrix.

**Owner:** Control systems engineer

**NIST Alignment:** SP 800-82r3; FIPS 203/204 where applicable

## 06 Map OT certificates and trust stores.

**Area:** PKI

**Safety / Operational Consideration:** Include historians, gateways, HMIs, engineering workstations, remote access, OPC UA, web interfaces, and device certificates.

**Evidence / Artifact:** OT certificate inventory.

**Owner:** PKI / OT engineering

**NIST Alignment:** SP 800-82r3; FIPS 204/205

## 07 Request OT vendor PQC roadmap and support windows.

**Area:** Vendor

**Safety / Operational Consideration:** Confirm firmware/software upgrade paths, protocol support, certificate/key size limits, hardware constraints, and validation process.

**Evidence / Artifact:** Vendor responses and support statement.

**Owner:** Vendor management / OT owner

**NIST Alignment:** SP 800-161r1; CISA OT PQC considerations

## 08 Build lab/digital twin test path for PQC-impacting changes.

**Area:** Testing

**Safety / Operational Consideration:** Never test unvalidated cryptographic changes directly on production control systems. Use representative devices where feasible.

**Evidence / Artifact:** Lab plan, test assets, acceptance criteria.

**Owner:** OT engineering / lab lead

**NIST Alignment:** SP 800-82r3 safety/reliability focus

## 09 Validate performance and packet-size impact.

**Area:** Testing

**Safety / Operational Consideration:** Assess latency, CPU/memory, handshake failures, fragmentation, certificate chain size, and recovery behavior on constrained devices.

**Evidence / Artifact:** Test results and performance baseline.

**Owner:** OT engineering / QA

**NIST Alignment:** NIST CSWP 39 resource/performance considerations

## 10 Define controls for non-upgradeable assets.

**Area:** Compensating Controls

**Safety / Operational Consideration:** Use segmentation, unidirectional gateways where appropriate, jump hosts, strict remote access, monitoring, lifecycle replacement, and accepted-risk expiration.

**Evidence / Artifact:** Compensating control plan.

**Owner:** OT security / risk owner

**NIST Alignment:** SP 800-82r3; SP 800-53 SC/AC/RA

## 11 Update OT monitoring and incident response.

**Area:** Operations

**Safety / Operational Consideration:** Ensure new or changed crypto protocols/certificates are visible in monitoring and runbooks; document emergency rollback.

**Evidence / Artifact:** Runbooks, alert rules, tabletop results.

**Owner:** SOC / OT operations

**NIST Alignment:** CSF Detect/Respond/Recover; SP 800-82r3

## 12 Integrate PQC readiness into capital planning.

**Area:** Lifecycle

**Safety / Operational Consideration:** Use asset refresh cycles to replace non-agile devices, unsupported firmware, and appliances with fixed cryptographic parameters.

**Evidence / Artifact:** Capital plan, lifecycle roadmap.

**Owner:** Operations / asset management

**NIST Alignment:** SP 800-82r3; NIST CSWP 39

## 13 Verify protected backups and restoration.

**Area:** Recovery

**Safety / Operational Consideration:** Restore controller logic, configuration, firmware, trust stores and recoverable keys in a representative lab. Confirm permissions, integrity and restoration time.

**Evidence / Artifact:** Backup scope, restore proof and approved recovery objectives.

**Owner:** OT engineering / operations

**NIST Alignment:** N15 SP 1339 final; N13 final

**Status:** Not assessed

**Notes:** Backup existence alone does not demonstrate recovery.

## 14 Find immutable boot and firmware verifiers.

**Area:** Trust anchors

**Safety / Operational Consideration:** Record hardware/ROM trust anchors, signed update chains, device provisioning and verification lifetime. Escalate replacement needs early.

**Evidence / Artifact:** Verifier dependency inventory and funded replacement plan.

**Owner:** OT asset owner / supplier

**NIST Alignment:** N06 draft; N09/N13 final

**Status:** Not assessed

**Notes:** Long-lived authenticity risks differ from HNDL confidentiality.

## 15 Test constrained and adverse conditions.

**Area:** Performance

**Safety / Operational Consideration:** Measure scan-cycle latency/jitter, CPU/memory, bandwidth, packet limits, malformed/oversize handling, reconnect, failover and resource exhaustion.

**Evidence / Artifact:** Vendor-supported lab tests, safety acceptance and thresholds.

**Owner:** OT lab / safety owner

**NIST Alignment:** N04/N05/N13 final

**Status:** Not assessed

**Notes:** Keep tests out of production until MOC and safety approval.

## 16 Record downstream gaps after gateways.

**Area:** Architecture

**Safety / Operational Consideration:** Map zones/conduits, plaintext or legacy protocols and every termination boundary. Record residual risk and scope of gateway protection.

**Evidence / Artifact:** Path diagram and time-bound exception evidence.

**Owner:** OT network / risk owner

**NIST Alignment:** N13 final

**Status:** Not assessed

**Notes:** Segmentation and a PQC gateway do not convert downstream legacy crypto.

## 17 Rehearse safe rollback and manual operation.

**Area:** Operations

**Safety / Operational Consideration:** Test approved restore/failover, safe state, manual operation, replacement equipment and emergency communication within operational limits.

**Evidence / Artifact:** Exercise report, named decision authority and recovery acceptance.

**Owner:** Operations / safety / SOC

**NIST Alignment:** N13/N15 final

**Status:** Not assessed

**Notes:** Record maintenance window, downtime tolerance and owner approval.

## 18 Review the 2026 OT draft without calling it final.

**Area:** Standards watch

**Safety / Operational Consideration:** Track SP 800-82 Rev.4 draft changes for asset management, monitoring, cloud/IIoT convergence and management systems. Maintain Rev.3 final baseline until superseded.

**Evidence / Artifact:** Dated standards review and approved gap decisions.

**Owner:** OT governance

**NIST Alignment:** N14 initial public draft 2026-09-21; N13 final

**Status:** Not assessed

**Notes:** Comments due 2026-11-30. Draft changes are not final requirements.

# References

Official sources, publication status, and applicability recorded in the workbook. Recheck draft status, errata, and implementation guidance before a production decision.

## **01 FIPS 203 ML-KEM; key establishment, parameter sets 512/768/1024; monitor potential errata**

**Publication Status / Scope:** Final 2024-08-13; notice 2025-11-17

**Official URL:** <https://csrc.nist.gov/pubs/fips/203/final>

**Source ID:** N01

**Checked Date:** 2026-09-29

## **02 FIPS 204 ML-DSA; signature generation and verification; monitor potential errata**

**Publication Status / Scope:** Final 2024-08-13; notice 2026-07-31

**Official URL:** <https://csrc.nist.gov/pubs/fips/204/final>

**Source ID:** N02

**Checked Date:** 2026-09-29

## **03 FIPS 205 SLH-DSA; stateless hash-based signatures**

**Publication Status / Scope:** Final 2024-08-13

**Official URL:** <https://csrc.nist.gov/pubs/fips/205/final>

**Source ID:** N03

**Checked Date:** 2026-09-29

## **04 SP 800-227; conforming KEM implementation, randomness, input validation, key destruction, authentication, key derivation/confirmation and multi-algorithm construction**

**Publication Status / Scope:** Final 2025-09-18

**Official URL:** <https://csrc.nist.gov/pubs/sp/800/227/final>

**Source ID:** N04

**Checked Date:** 2026-09-29

## **05 CSWP 39-upd1; governance, inventory, risk-prioritized agility, negotiation, supply chains, APIs, continuous improvement**

**Publication Status / Scope:** Final 2025-12-19; updated 2026-06-29

**Official URL:** <https://csrc.nist.gov/pubs/cswp/39/upd1/considerations-for-achieving-crypto-agility/final>

**Source ID:** N05

**Checked Date:** 2026-09-29

## **06 IR 8547; HNDL, signature and code-verification distinctions, draft transition tables**

**Publication Status / Scope:** Initial public draft 2024-11-12

**Official URL:** <https://csrc.nist.gov/pubs/ir/8547/ipd>

**Source ID:** N06

**Checked Date:** 2026-09-29

## **07 NCCoE Migration to PQC; discovery/inventory and interoperability workstreams; evolving output model**

**Publication Status / Scope:** Active project; preliminary documents

**Official URL:** <https://www.nccoe.nist.gov/applied-cryptography/migration-to-pqc>

**Source ID:** N07

**Checked Date:** 2026-09-29

## **08 SP 1800-38 B/C; discovery and interoperability/performance demonstration**

**Publication Status / Scope:** Initial preliminary draft 2023-12-19

**Official URL:** [https://csrc.nist.gov/pubs/sp/1800/38/iprd-\(1\)](https://csrc.nist.gov/pubs/sp/1800/38/iprd-(1))

**Source ID:** N08

**Checked Date:** 2026-09-29

## **09 SP 800-57 Part 1 Rev. 5; key management baseline**

**Publication Status / Scope:** Final 2020-05-04

**Official URL:** <https://csrc.nist.gov/pubs/sp/800/57/pt1/r5/final>

**Source ID:** N09

**Checked Date:** 2026-09-29

## **10 SP 800-57 Part 1 Rev. 6; PQC-aware key management and storage proposals**

**Publication Status / Scope:** Initial public draft 2025-12-05

**Official URL:** <https://csrc.nist.gov/pubs/sp/800/57/pt1/r6/ipd>

**Source ID:** N10

**Checked Date:** 2026-09-29

## **11 SP 800-131A Rev. 2; current final algorithm transition baseline**

**Publication Status / Scope:** Final 2019-03-21

**Official URL:** <https://csrc.nist.gov/pubs/sp/800/131/a/r2/final>

**Source ID:** N11

**Checked Date:** 2026-09-29

## **12 SP 800-131A Rev. 3; proposed algorithm/key-strength transitions**

**Publication Status / Scope:** Initial public draft 2024-10-21

**Official URL:** <https://csrc.nist.gov/pubs/sp/800/131/a/r3/ipd>

**Source ID:** N12

**Checked Date:** 2026-09-29

### **13 SP 800-82 Rev. 3; OT performance, reliability, safety and controls**

**Publication Status / Scope:** Final 2023-09-28

**Official URL:** <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

**Source ID:** N13

**Checked Date:** 2026-09-29

### **14 SP 800-82 Rev. 4; CSF2 alignment, asset management, monitoring, OT/IIoT/cloud, system-management security**

**Publication Status / Scope:** Initial public draft 2026-09-21

**Official URL:** <https://csrc.nist.gov/pubs/sp/800/82/r4/ipd>

**Source ID:** N14

**Checked Date:** 2026-09-29

### **15 SP 1339; backups in OT change and recovery practice**

**Publication Status / Scope:** Final 2026-06-17

**Official URL:** <https://csrc.nist.gov/pubs/sp/1339/final>

**Source ID:** N15

**Checked Date:** 2026-09-29

### **16 IR 8587; signed tokens/assertions, key lifecycle, verification, workload/automated-workload token use, Section 7.7 PQC interoperability constraints**

**Publication Status / Scope:** Final 2026-09-15

**Official URL:** <https://csrc.nist.gov/pubs/ir/8587/final>

**Source ID:** N16

**Checked Date:** 2026-09-29

### **17 SP 800-228-upd1; API lifecycle risks and controls**

**Publication Status / Scope:** Final; original 2025-06-27, updated 2026-03-13

**Official URL:** <https://csrc.nist.gov/pubs/sp/800/228/upd1/final>

**Source ID:** N17

**Checked Date:** 2026-09-29

### **18 CSWP 48; mapping PQC discovery and interoperability capabilities to risk frameworks**

**Publication Status / Scope:** Initial public draft 2025-09-18

**Official URL:** <https://csrc.nist.gov/pubs/cswp/48/mapping-migration-to-pqc-project-capabilities-to-r/ipd>

**Source ID:** N18

**Checked Date:** 2026-09-29

## 19 NIST PQC project; current standardized/selected algorithm status

**Publication Status / Scope:** Project updated 2026-08-05

**Official URL:** <https://csrc.nist.gov/projects/post-quantum-cryptography>

**Source ID:** N19

**Checked Date:** 2026-09-29

## 20 HQC selection announcement; diversification backup, continue current migration, expected 2027 finalization

**Publication Status / Scope:** Selection announcement 2025-03-11, forecast not approval

**Official URL:**

<https://www.nist.gov/news-events/news/2025/03/nist-selects-hqc-fifth-algorithm-post-quantum-encryption>

**Source ID:** N20

**Checked Date:** 2026-09-29

## 21 Additional digital signature schemes; Round3 announced May14,2026

**Publication Status / Scope:** Evaluation candidates, not standards

**Official URL:** <https://csrc.nist.gov/projects/pqc-dig-sig>

**Source ID:** N21

**Checked Date:** 2026-09-29

## 22 SP 800-204A; service-mesh security architecture

**Publication Status / Scope:** Final 2020-05-27

**Official URL:** <https://csrc.nist.gov/pubs/sp/800/204/a/final>

**Source ID:** N22

**Checked Date:** 2026-09-29

## 23 SP 800-204B; service-mesh authentication/authorization

**Publication Status / Scope:** Final 2021-08

**Official URL:** <https://csrc.nist.gov/pubs/sp/800/204/b/final>

**Source ID:** N23

**Checked Date:** 2026-09-29

## 24 CAVP program; algorithm validation is prerequisite to, not replacement for, module validation

**Publication Status / Scope:** Official program; updated 2026-09-22

**Official URL:** <https://csrc.nist.gov/projects/cryptographic-algorithm-validation-program>

**Source ID:** N24

**Checked Date:** 2026-09-29

## **25 CMVP FAQ; module/version/operating-environment verification, protocol/product validation boundary, FIPS140-2 active-list transition**

**Publication Status / Scope:** Official FAQ checked 2026-09-29

**Official URL:** <https://csrc.nist.gov/Projects/cryptographic-module-validation-program/faqs>

**Source ID:** N25

**Checked Date:** 2026-09-29

## **26 CMVP validated modules; inspect certificate/security policy and active/historical/revoked status**

**Publication Status / Scope:** Live certificate registry checked 2026-09-29

**Official URL:** <https://csrc.nist.gov/projects/cryptographic-module-validation-program/validated-modules>

**Source ID:** N26

**Checked Date:** 2026-09-29

## **27 CSWP 29 / CSF 2.0; cybersecurity outcome framework, not a prescriptive PQC implementation standard**

**Publication Status / Scope:** Final 2024-02-26

**Official URL:** <https://csrc.nist.gov/pubs/cswp/29/the-nist-cybersecurity-framework-csf-20/final>

**Source ID:** N27

**Checked Date:** 2026-09-29

## **28 SP 800-53 Rev. 5; current control catalog is Release 5.2.0 referenced by publication-page planning note**

**Publication Status / Scope:** Final; base 2020-09/update 2020-12-10; minor Release 5.2.0 finalized 2025-08-27

**Official URL:** <https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final>

**Source ID:** N28

**Checked Date:** 2026-09-29

## **29 SP 800-161 Rev. 1-upd1; supplier risk and dependency governance**

**Publication Status / Scope:** Final; original 2022-05, update 2024-11-01

**Official URL:** <https://csrc.nist.gov/pubs/sp/800/161/r1/upd1/final>

**Source ID:** N29

**Checked Date:** 2026-09-29

## **30 SP 800-230; additional limited-signature SLH-DSA parameter sets, software/firmware/certificate watch item**

**Publication Status / Scope:** Initial public draft 2026-04-13

**Official URL:** <https://csrc.nist.gov/pubs/sp/800/230/ipd>

**Source ID:** N30

**Checked Date:** 2026-09-29

## 31 SP 800-133 Rev. 3; proposed PQC key generation, seed expansion, KEM and HSM considerations

**Publication Status / Scope:** Initial public draft 2026-04-17

**Official URL:** <https://csrc.nist.gov/pubs/sp/800/133/r3/ipd>

**Source ID:** N31

**Checked Date:** 2026-09-29

## 32 CISA Migration to Post-Quantum Cryptography

**Publication Status / Scope:** Context reference from v1.0. Recheck current status before reliance. Quantum-readiness roadmap and inventory considerations.

**Official URL:**

<https://www.cisa.gov/resources-tools/resources/quantum-readiness-migration-post-quantum-cryptography>

**Source ID:** L11

## 33 CISA Post-Quantum Considerations for Operational Technology

**Publication Status / Scope:** Context reference from v1.0. Recheck current status before reliance. OT-specific PQC migration considerations.

**Official URL:** <https://www.cisa.gov/resources-tools/resources/post-quantum-considerations-operational-technology>

**Source ID:** L12

## 34 CISA Product Categories for PQC Adoption

**Publication Status / Scope:** Context reference from v1.0. Recheck current status before reliance. Product categories that currently support or are expected to support PQC standards.

**Official URL:** <https://www.cisa.gov/resources-tools/resources/product-categories-technologies-use-post-quantum-cryptography-standards>

**Source ID:** L13

# Roadmap

Read this section alongside the corresponding worksheet in the editable workbook. Record decisions and evidence there.

## 01 Reconcile discovery, crypto component owners, long-lived confidentiality/signing risks, supplier evidence and immovable trust anchors.

**Period:** 2026 baseline

**Exit Evidence:** Coverage gaps, risk-approved backlog and current supplier statements.

**Owner:** Unassigned

**Dependencies / Risks:** Inventory access, business owners and supplier engagement.

**Status:** Not assessed

**Source IDs:** N05 final; N06 draft; N07 project

**Exception / Review:** Set organization-specific completion date.

## 02 2027 implementation

**Local Planning Action:** Pilot supported high-risk paths, PKI/identity interoperability and OT recovery. Add agility and versioned support commitments to procurement.

**Exit Evidence:** Measured end-to-end security/performance tests, rollback and approval.

**Owner:** Unassigned

**Dependencies / Risks:** Protocol maturity, vendor versions, HSM/CA/validator support.

**Status:** Not assessed

**Source IDs:** N01-N05,N15-N17 final

**Exception / Review:** HQC expected-2027 announcement is a forecast, not approval.

## 03 Expand successful migrations and retire avoidable legacy dependencies. Recheck proposed 112-bit transitions against then-current NIST publications.

**Period:** 2028-2030 scale

**Exit Evidence:** Observed deployment, funded replacement and exception burn-down.

**Owner:** Unassigned

**Dependencies / Risks:** Risk-based sequencing and lifecycle windows.

**Status:** Not assessed

**Source IDs:** N05 final; N06 draft

**Exception / Review:** No blanket 2030 deadline for all RSA/ECC.

## 04 2031-2035 closure

**Local Planning Action:** Close long-lived/vendor-limited gaps early enough for applicable final policies. Remove obsolete fallback paths and decommission vulnerable components.

**Exit Evidence:** Independent acceptance, closure evidence and time-bound exceptions.

**Owner:** Unassigned

**Dependencies / Risks:** Draft IR 8547 proposes disallowance after 2035 for listed vulnerable uses.

**Status:** Not assessed

**Source IDs:** N06 draft; N09/N11 final

**Exception / Review:** Treat draft dates as planning signals; confirm applicable policy.

## 05 Refresh inventory and tests; review new algorithms, cryptanalysis, errata, vendor/module status and retention/verification obligations.

**Period:** 2036+ ongoing

**Exit Evidence:** Dated review, incident exercises and refreshed dependency evidence.

**Owner:** Unassigned

**Dependencies / Risks:** Standards changes and remaining archived verification needs.

**Status:** Not assessed

**Source IDs:** N05/N09 final; N19-N21 watch

**Exception / Review:** Quarterly and event-triggered cadence is local guidance.

# Threat Register

Threat scenarios are planning inputs. Assign an accountable owner, record affected dependencies, and retain the evidence needed to test each treatment.

## 01 Harvest now, decrypt later

**Threat ID:** T01

**Affected Function / Scope:** Long-lived confidential traffic, envelope keys, backups and archives.

**Required Assessment / Evidence:** Confidentiality lifetime, capture exposure, key-establishment/wrapping dependency and migration lead time.

**Suggested Treatment:** Prioritize supported protective migration and minimize exposure/retention.

**Owner:** Unassigned

**Status:** Not assessed

**Source IDs:** N06 draft; N04/N09 final

## 02 Signature forgery and impersonation

**Threat ID:** T02

**Affected Function / Scope:** PKI, tokens, software/firmware signing, boot roots and archived validation.

**Required Assessment / Evidence:** Verification lifetime, trust anchors, validators and immutable hardware constraints.

**Suggested Treatment:** Plan compatible signing/trust rollover and preserve required verification evidence.

**Owner:** Unassigned

**Status:** Not assessed

**Source IDs:** N02/N03/N09/N16 final; N06 draft

## 03 Downgrade or unsafe fallback

**Threat ID:** T03

**Affected Function / Scope:** Negotiated/hybrid protocols and emergency rollback.

**Required Assessment / Evidence:** Policy allowlists, full construction, authentication and negative negotiation tests.

**Suggested Treatment:** Use supported profiles, authenticated negotiation and approved time-bound rollback.

**Owner:** Unassigned

**Status:** Not assessed

**Source IDs:** N04/N05 final

## 04 Incomplete discovery or false readiness

**Threat ID:** T04

**Affected Function / Scope:** Libraries, certificates, hidden proxies, plaintext legs and dependencies.

**Required Assessment / Evidence:** Observed/configured/vendor evidence, provenance, confidence and coverage gaps.

**Suggested Treatment:** Continuous reconciliation, owner assignment and explicit unknown states.

**Owner:** Unassigned

**Status:** Not assessed

**Source IDs:** N05 final; N07 project; N08 preliminary draft

## 05 Implementation flaws or key compromise

**Threat ID:** T05

**Affected Function / Scope:** Randomness, input validation, side channels, secret handling and recovery.

**Required Assessment / Evidence:** Version/errata evidence, secure key lifecycle, test vectors and implementation review.

**Suggested Treatment:** Patch approved versions; validate inputs and protect/destroy keys per lifecycle.

**Owner:** Unassigned

**Status:** Not assessed

**Source IDs:** N01-N04/N09 final

## 06 Availability or interoperability failure

**Threat ID:** T06

**Affected Function / Scope:** Large handshakes, certificates, tokens, parsers, networks and constrained OT.

**Required Assessment / Evidence:** Measured latency, jitter, memory/CPU, MTU, limits, retries, failover and restore.

**Suggested Treatment:** Representative end-to-end lab/canary validation with service/safety acceptance.

**Owner:** Unassigned

**Status:** Not assessed

**Source IDs:** N04/N05/N13/N15/N16 final

## 07 Supplier or lifecycle blockers

**Threat ID:** T07

**Affected Function / Scope:** EOL devices, unsupported firmware, HSM/KMS/CA and immutable verifiers.

**Required Assessment / Evidence:** Dated version support, product EOL, replacement lead time and dependency map.

**Suggested Treatment:** Contractual commitments, funded replacement and expiring accepted risk.

**Owner:** Unassigned

**Status:** Not assessed

**Source IDs:** N05/N13 final

## 08 Misstated standards or validation status

**Threat ID:** T08

**Affected Function / Scope:** Drafts, candidates, marketing claims and FIPS module scope.

**Required Assessment / Evidence:** Final/draft register, algorithm versus module certificate, exact version and environment.

**Suggested Treatment:** Recheck authoritative status; do not declare compliant from algorithm support alone.

**Owner:** Unassigned

**Status:** Not assessed

**Source IDs:** N01-N05; N06/N10/N12/N14 drafts; N19-N21 watch

## 09 Recovery failure or lingering legacy verification

**Threat ID:** T09

**Affected Function / Scope:** Backup keys/configuration, trust stores, archived signatures and rollback.

**Required Assessment / Evidence:** Tested restore, retention/verification obligation and approved retirement plan.

**Suggested Treatment:** Protect recovery material, rehearse restore and decommission expired paths.

**Owner:** Unassigned

**Status:** Not assessed

**Source IDs:** N09/N13/N15 final